

** Note: Payloads are shown in JSON/Plain-text format for clarity, but are transmitted as packed C-structs over LoRa to minimize time-on-air, battery usage, and collision probability.*

Node (Mota)
Client

Router
Coordinator

Wakes up. Prepares Payload.

DATA (0x20) / JOIN (0x02)
Payload includes **fcnt** $\leq$ **lastFCnt** (Reused/Old)

Anti-Replay Validation Failed!
Desynchronization or Replay Attack

CRYPTO_ERROR (0x24)
Payload: ControlData

Increment **crypto_err** counter
Forced Resynchronization:
State = STATE_START_JOIN

JOIN_REQUEST (0x02)
Payload: [Router: 10, ID: 50, fcnt = ++join_cnt]

Accepts new Join,
Resets **lastFCnt** Tracker